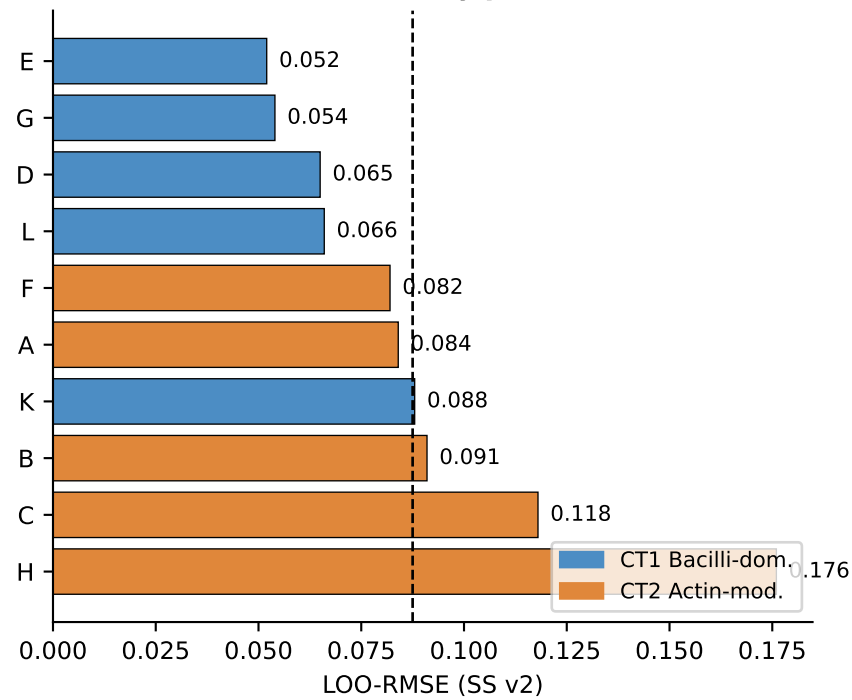
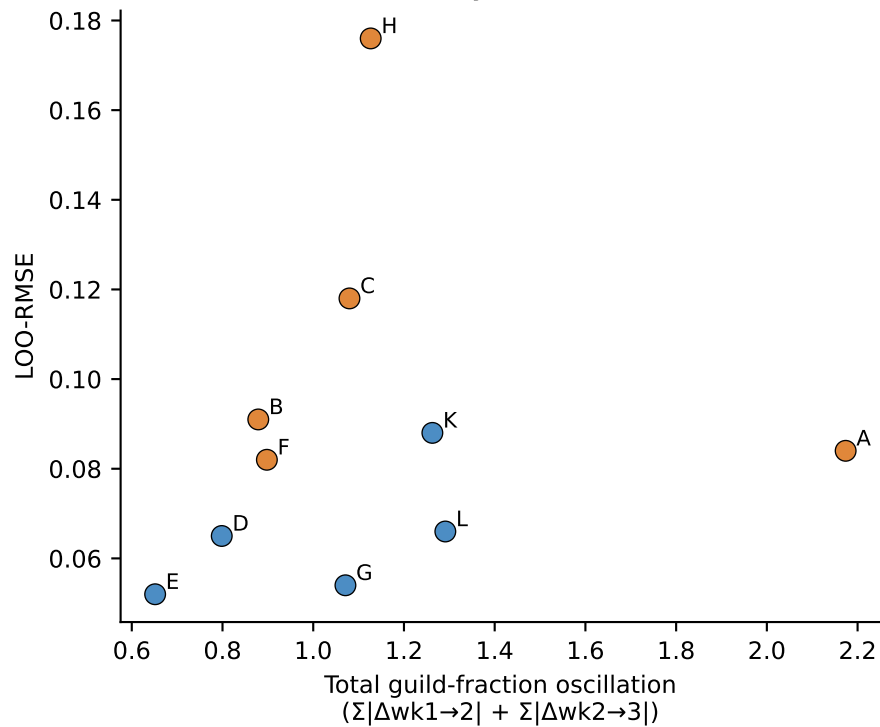


LOO-CV Failure Analysis — Patients H & C (SS v2 model)

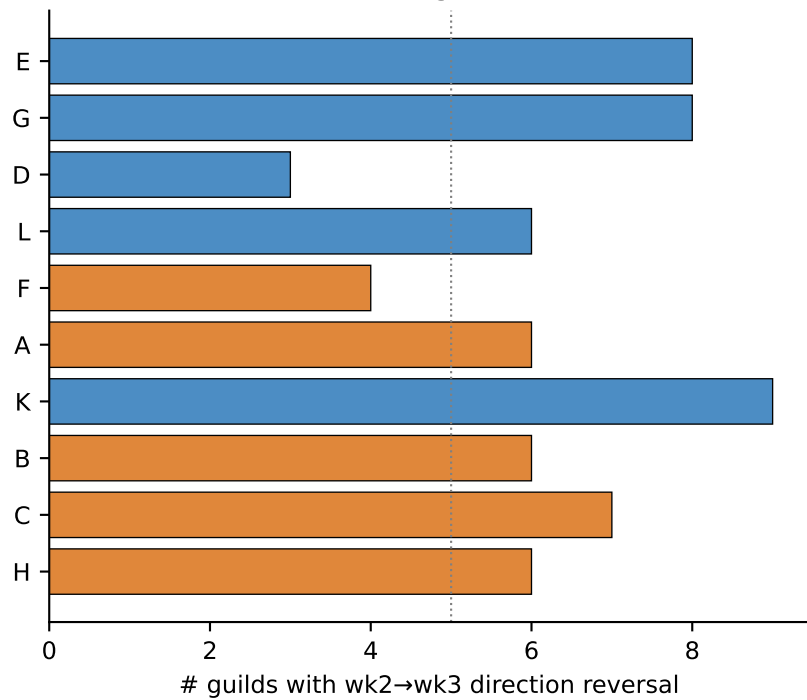
LOO-RMSE by patient



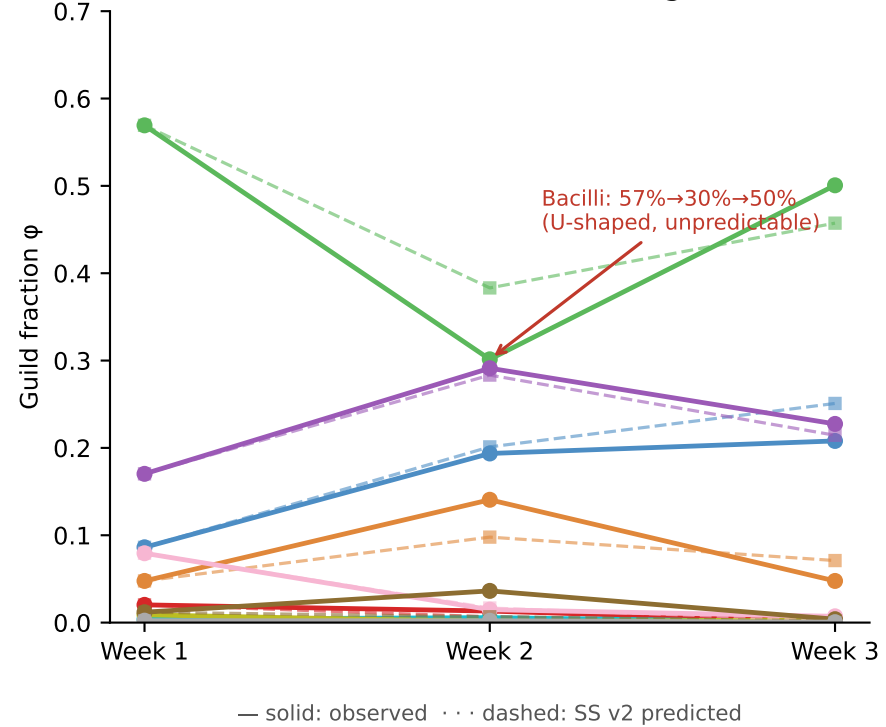
LOO-RMSE vs dynamics oscillation



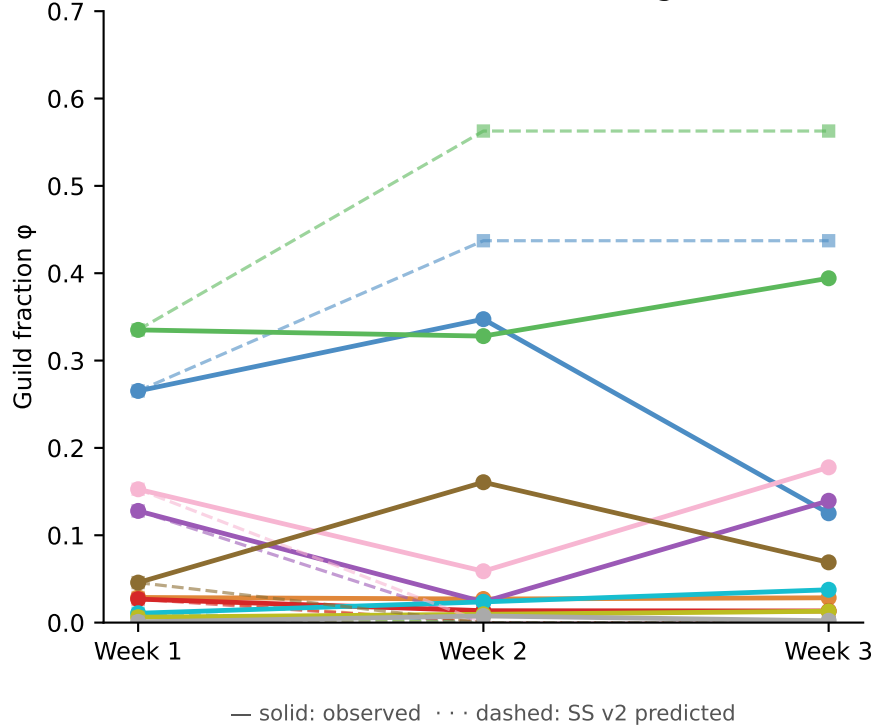
Non-monotone guild reversals



Patient H — LOO-RMSE=0.176, 6/10 guild reversals



Patient C — LOO-RMSE=0.118, 7/10 guild reversals



Root causes of high L00-RMSE

- Patient H (LOO-RMSE = 0.176)
Bacilli: 57% → 30% → 50% (U-shape)
8/10 guilds reverse direction wk2→wk3
→ Non-monotone transient dynamics
→ Possibly: external perturbation, measurement noise in wk2 sample, or multi-attractor basin crossing
- Patient C (LOO-RMSE = 0.118)
Negativicutes: 5% → 16% → 7%
Gammaproteobacteria: 15% → 6% → 18%
→ Transient peak/valley pattern
→ ODE predicts monotone convergence, misses transient overshoot
- Model limitation:
SS v2 predicts convergence to a single steady state; cannot capture reversals or transient overshoots within a 3-week window.

